Concise but significant* handbook to M5 statistics

Hotel Management School Maastricht ZUYD

* p = 0.01

What measurement level?

Variable
data

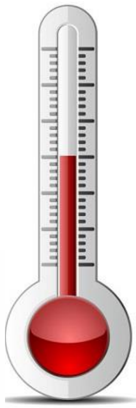
Categorical
<u>ABC</u>
Labels, making groups, distinct categories

Nominal
Categories are equal (= and ≠)

Ordinal
There is an order to the categories (> and <)

Continuous
<u>123</u>
Measured, scale, ranges, values

Interval
Equal distance between the values (+ and -)



Ratio
*There is an absolute 0 point (* and /)*

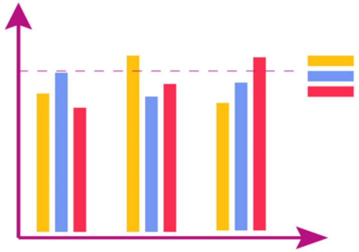
Which chart to use?

Variable
data

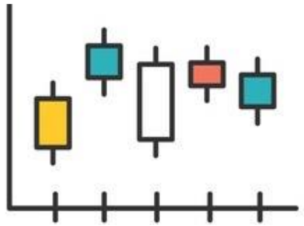
Categorical **ABC**
Labels, making groups, distinct categories

1 variable
Bar chart or pie chart

2 variables
Clustered or stacked bar chart



1 of each
Various options: Bar chart, error bar plot, box chart



Continuous
123
Measured, scale, ranges, values

1 variable
Histogram

2 variables
Scatterplot, or line graph (over time)

Making hypotheses

A research hypothesis is a concise statement about the expected result of an experiment or project. Two types exist:

- the ***null hypothesis*** H_0 Assumed to be factual, normal, ´what there is´ in the current world.
- the ***alternative* hypothesis** H_1 / H_a What could or might be, what we look for out of the ordinary.

→ Always formulate H1 first, then add H_0 as the opposite.

Example: A researcher tests the effectiveness of a new medication. In an "alternative" world, the new medication would have some effect, and that is also what we hope and expect, so H_1 is: "the medication makes a difference":

H_1 = the medication has an effect ALSO $\mu_{test} \neq \mu_{control}$

Normally, however, a random pill does not do a lot, so the "normal reality" of H_0 is: the medication has no effect. In that case, we do not expect any difference between the means (μ's) of the test group and the control group:

H_0 = the medication has no effect ALSO $\mu_{test} = \mu_{control}$

The hypotheses above do not have any direction. They simply deal with a possible difference in μ´s. This test is **two-sided**: any result could be more or less, better or worse. However, we could also formulate a hypothesis with a direction, in which case we could say: the drug ***improves*** health. Now, we test **one-sided**: the result of the test group should be higher/better.

H_1 = the drug has a positive effect ALSO $\mu_{test} > \mu_{control}$

Hypotheses, $\bar{x}$, μ, p, and distributior

Statistical tests – simply put – calculate how likely it would be to find a result like the one in your sample (or an even more extreme result) if H_0 were true. This probability is the p-value.

Why? If H_1 says that there really is a difference or effect, we want the difference found in our sample to be large enough that it would be unlikely to occur just by chance if H_0 were true.

Normally, we use a critical p-value (α) of 0.05. If $p < 0.05$, our result would be sufficiently unlikely under H_0, so we reject H_0 in favour of H_1. If $p \geq 0.05$, we do not reject H_0 because there is not enough evidence for H_1.

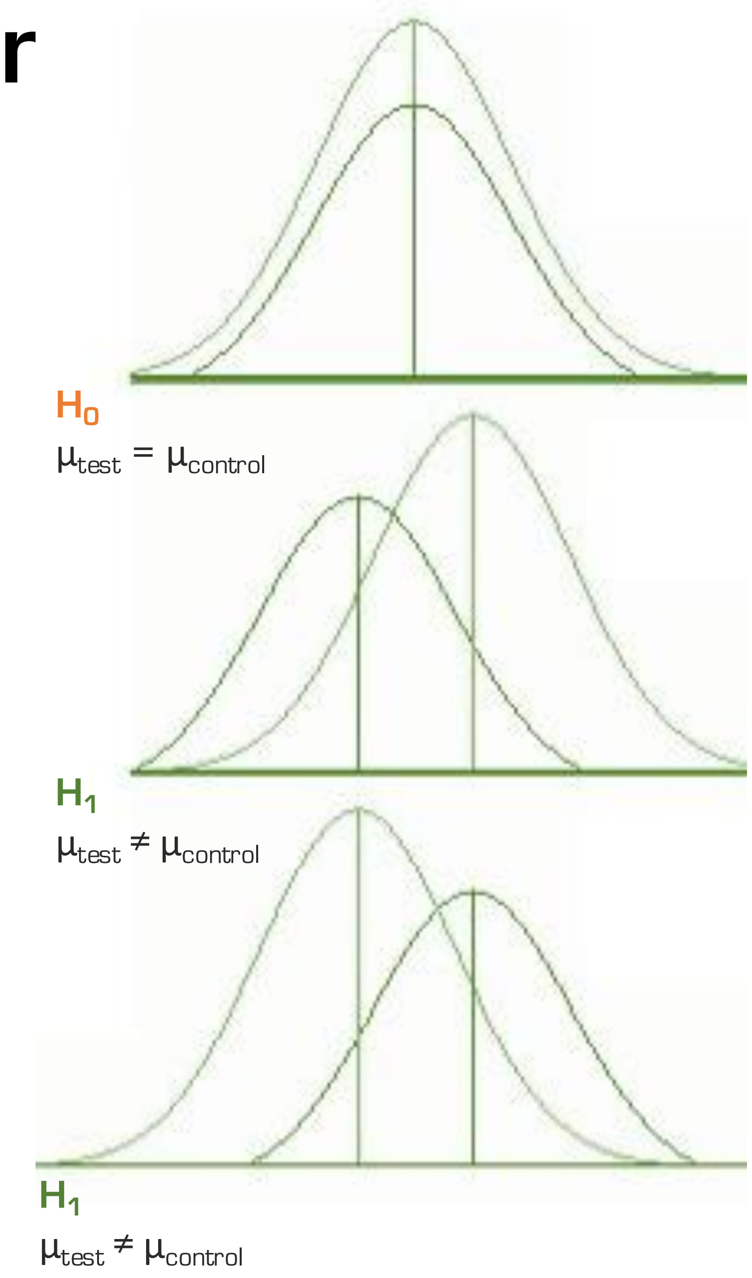
A 95% confidence interval gives us a range of plausible values for the population parameter we are estimating. In many common two-sided tests, if the value predicted by H_0 falls outside the 95% confidence interval, the result is significant at the 0.05 level.

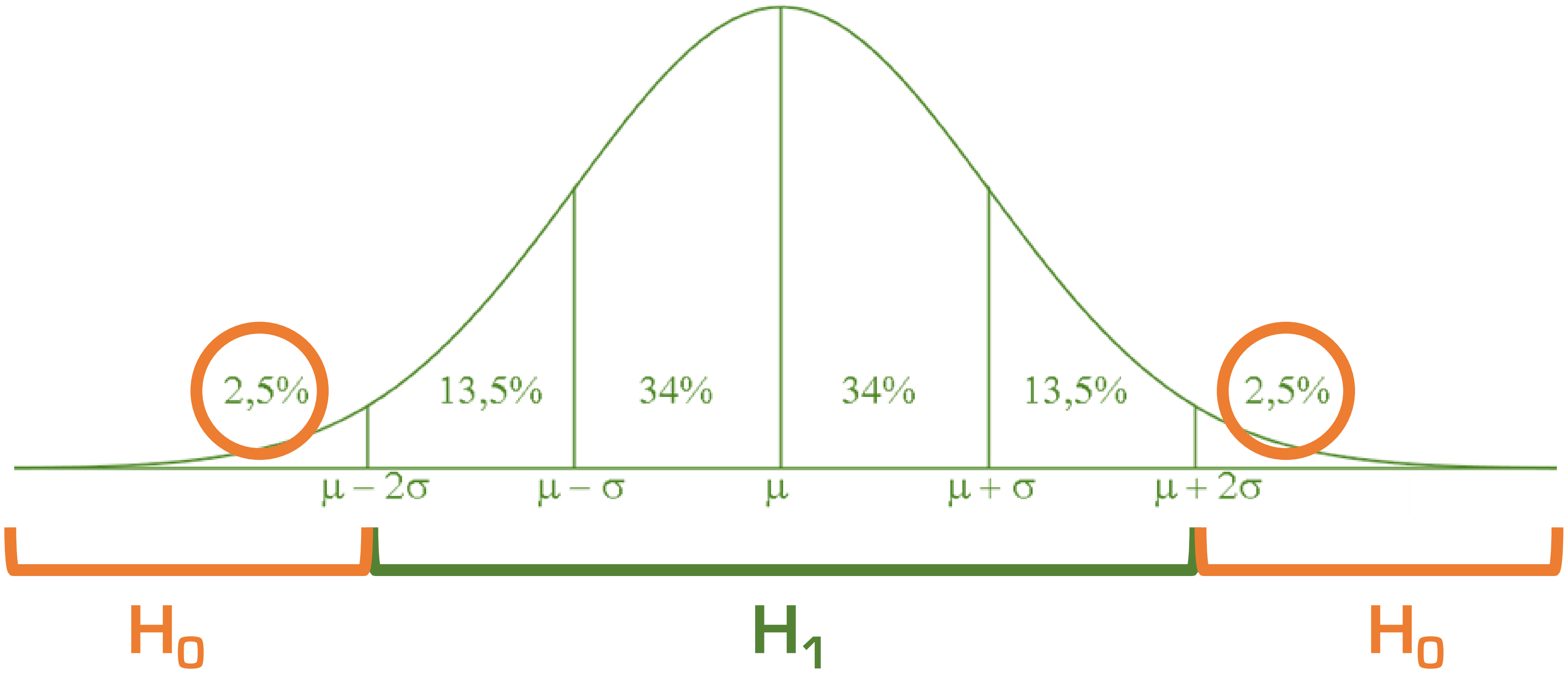
NOTES:

A. When testing hypotheses about means, we use μ (mu) for the population mean. When testing hypotheses about proportions (parts/shares), we use π (pi) for the population proportion. Hypotheses are therefore formulated using μ or π.

B. μ is the true mean of the whole population, but because populations are often too large to measure completely, μ is usually unknown. We therefore take a sample and calculate $\bar{x}$ (x-bar), the mean of that sample, which we use to estimate μ.

If $p < \alpha$ we reject H_0 in favour of H_1. If $p \geq \alpha$, we do not reject H_0.





If the resulted probability p is below the chosen critical value p (e.g. 0.05), then
there are limited cases in the H_0 region and we can reject H_0
&
we assume there are enough cases in the H_1 region to assume a difference in μ´s and support H_1, so we accept H_1.

! If you test one-sided, then the test´s p-value has to be divided by 2 (you can´t ´use´ the other side of the distribution). In Jamovi you can tick boxes for testing either two-sided or one-sided, in SPSS the results table reports both.

Types of error

If you use statistics to make decisions on hypotheses, the chance exists that you draw false conclusions. Options here are:

Type 1-error, or a false positive

The rejection of the H_0 when it is actually true and should be assumed

Thus also meaning: accepting H_1 while you shouldn´t.

Type 2-error, or a false negative

Not rejecting H_0 while it is actually false and should be rejected

Thus also meaning: not accepting H_1 while you actually should.

Type 3-error, for the enthusiasts

Rejecting H_0 because the numbers say so, but based on wrong reasoning/methodology

E.g.1 Concluding ´*the drug* ***improves*** *health*´ with two-tailed testing

E.g.2 The drug has a positive effect and people feel better, but it doesn´t concern the theoretical effect and illness that was studied

Validity and Reliability

Validity refers to the accuracy of a measure: do you measure what is supposed to be measure?

e.g. A new questionnaire is used and the results align with those of previously used questionnaires on the same topic among a comparable population 👍

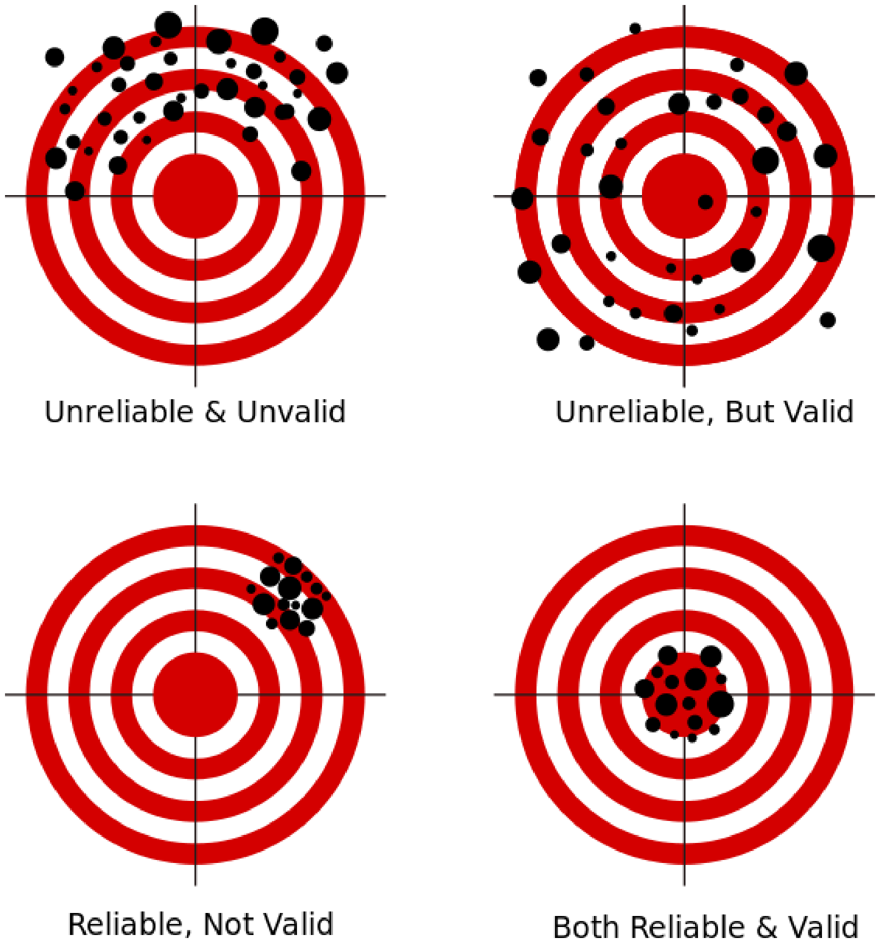
e.g. A thermometer is placed next to the door in a controlled laboratory fridge 👎

Reliability refers to the consistency of a measure and data: can results be reproduced under the same conditions, and if the dataset size N increases, does the outcome of the test increasingly approaches the correct outcome?

e.g. A respondent answers ´yes´ to *Do you eat meat?* and ´no´ to *Are you a vegetarian?* 👍

e.g.2 The more respondents (higher N), the more significant differences are found, which strengthens the outcome

→ Internal consistency is usually expressed with Cronbach's alpha α



Validity

Face Validity

Expert judgement

Does it make sense at first sight, also to participants?

Content Validity

Completeness

Does the test cover everything it should?

Criterion Validity

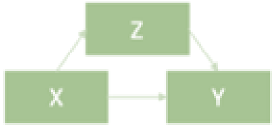
Does one measurement corrrelate with another?

Does a test accurately predict or correlate with the criterion outcome?

Construct Validity

Am I measuring what I want to measure?

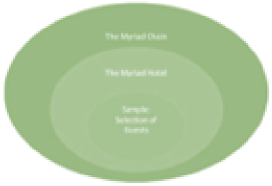
Does a set of indicators represent a concept that is not directly measurable?



Internal Validity

Causality X -> Y without Z influencing the relation

Does the study design, conduct, and analysis answer the research question without bias?



External validity

Can you generalise your results?

Can the study findings can be generalized to other contexts?

Reliability

- **Sample Size (N)**
- **Test-retest**

- **Standardization**
- **Pilots**
- **Peer Feedback**
- **Reporting and justification**

Quantitative

- **Intersubjectivity**
- **Triangulation**
- **Iteration**

Qualitative

Sampling

→ **Set criteria for your target population, e.g. age limits or country**
→ **Get a sample of subjects* that is as <u>random</u> and <u>representative</u> as possible.**

Random: the word says it all. Don´t cherry-pick or go around asking only friends and family.
Representative: The sample should contain the best mix of characteristics *within your target population*.

PROBABILITY SAMPLING:	based on the principle of randomization: random selection or chance within population or subpopulation
NON-PROBABILITY SAMPLING:	Selection is influenced by subjective boundaries and/or convenient picking from population or subpopulation

*** Research units are often people, but they can also be animals or objects, for example.**

Sampling: options

PROBABILITY SAMPLING OPTIONS

- (SIMPLE) RANDOM — Any subject within the target population (random) from database/ list
- STRATIFIED — Divide the population into subgroups (strata) based on a relevant characteristic, then randomly sample people from every subgroup, normally keeping the same proportions as the population.
  e.g. A university has 50% Dutch, 20% German, 20% Belgian, and 10% French students. For a sample of 1,000 students, you randomly select 500 Dutch, 200 German, 200 Belgian, and 100 French students.
- CLUSTERED — Divide the population into natural groups (clusters), randomly select some of those groups, and then survey everyone or a random sample of people within the selected clusters.
  e.g. You want to study language students across the Netherlands, but visiting every university and HBO is impractical. You treat individual language departments or schools as clusters, randomly choose several of them, and survey students within only those selected departments.
- SYSTEMATIC — Using some order or pattern to select, such as every 10^{th} guest

NON-PROBABILITY SAMPLING OPTIONS

- SNOWBALL — Any subject can select a next subject
- QUOTA — Divide the population into set subgroups and select a certain number of people from each subgroup, without random selection. You continue sampling until the required quota for each subgroup is reached.
  e.g. If a hotel has 80% guests staying in regular rooms and 20% staying in suites, and you want a sample of 50 guests, you select 40 guests from regular rooms and 10 guests from suites. You stop once those quotas have been reached.
- SELF-SELECTION — People put themselves forward for participation
- CONVENIENCE — Quick and convenient: any subject that you can find within a certain period or area. Very ´not random´.

What test to use?

- Checking relationships
  - Nominal / Ordinal *(Categorical)* → **Chi²** — *Expected counts ≥ 5 !!*
  - Ratio / Interval *(Continuous)* → **Correlation** — *Linear data only*
- Comparing means (μ´s)*
  - > 2 groups → **ANOVA**
  - 2 groups → **T-test**
    - One sample — *1 group vs test value*
    - Paired sample — *One group before vs after*
    - Independent sample — *2 separate groups*
    - Mann-Whitney U — *In case of violation of data assumption(s)*

*When comparing means, you always have to have one variable as your grouping or factor variable (to define groups), which is nominal or ordinal level. The other (test-) variable is measured at either interval or ratio level.

	Test	Data assumptions	Results
Checking Relationships	**Chi^2** 2 categorical variables	All expected counts ≥5 Independent measures	*p* for significance Cramer´s V for strength (0 to 1)
	Correlation 2 continuous variables	Linear data (scatterplot) No crazy outliers	*p* for significance Pearsons´s r for strength (-1 to 1)
Comparing Means *	**T-test** 1 continuous & 1 catagorical variable	Normally distributed data Equal variances (homogeneity)	*p* for significance Effect size: Cohen´s d (0 to 1) or Rank biserial correlation (-1 to 1)
	ANOVA 1 continuous & 1 catagorical variable **Multiple groups**	Equal variances (homogeneity)	*p* for significance Post-hoc for specific groups Groups means for specifications

* When comparing means, you always have to have one variable as your grouping or factor variable (to define groups), which is nominal or ordinal level (e.g. gender or age group). The other (test-) variable(s) is/are measured at continuous level.

CHI²

- The Chi² test, or χ^2 test, is used to see if there is a relationship between two categorical variables.
- Because categorical variables don´t have numerical measures and therefore cannot be analysed using means, Chi² works with frequencies to see if there are differences between observed outcomes (your measured counts) and expected counts.
- The expected outcomes for all cells should be at least 5, and are calculated using the formula (rowtotal x columntotal) / N, for example (28x32) /106 for cell **1-NO** in the table below
- You find the Chi² test in Jamovi under Analyses > Frequencies > Independent Samples χ^2 of association
- Jamovi provides a contingency table, the χ^2 test outcomes, and Cramer´s V (if selected under *Statistics*)

Contingency Tables

		healthplan		
qualification		No	Yes	Total
1	Observed	12	16	28
	Expected	8.45	19.55	28.00
2	Observed	18	27	45
	Expected	13.58	31.42	45.00
3	Observed	0	8	8
	Expected	2.42	5.58	8.00
4	Observed	2	18	20
	Expected	6.04	13.96	20.00
5	Observed	0	5	5
	Expected	1.51	3.49	5.00
Total	Observed	32	74	106
	Expected	32	74	106

χ^2 Tests

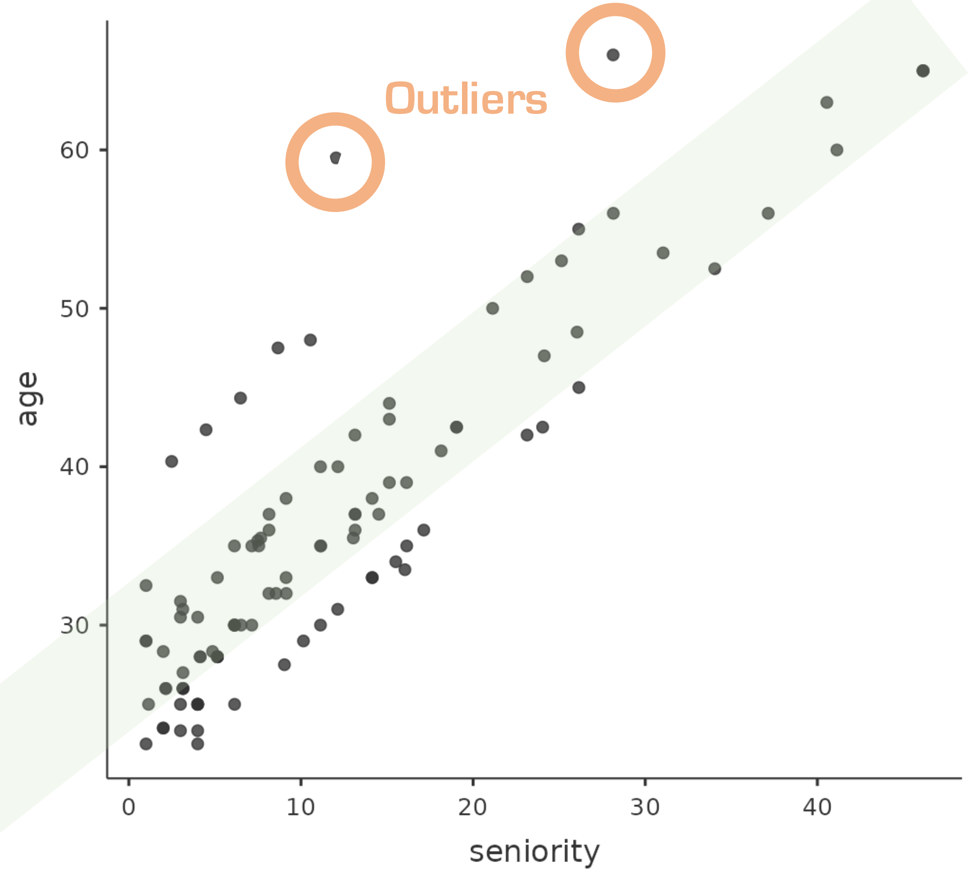
	Value	df	p
χ^2	13.7	4	0.008
N	106		

Nominal

	Value
Phi-coefficient	NaN
Cramer's V	0.359

CORRELATION

- The correlation test is used to see if there is a relationship between two continuous variables
- To check data assumptions, in Jamovi use Exploration > Scatterplot to see if data are reasonably linear or not
- You find the correlation test in Jamovi under Analyses > Regression > Correlation Matrix
- Jamovi provides a Correlation Matrix with the outcomes, among which Pearson´s r



Correlation Matrix

		age	seniority
age	Pearson's r	—	
	df	—	
	p-value	—	
seniority	Pearson's r	0.882	—
	df	104	—
	p-value	<.001	—

T-TEST

- The T-test is used compare the means (µ´s) of **two groups** for a continuous dependent variable, to see if there is a significant difference between the groups. There are 3 types of T-test.
    - **One sample t-test**. Used to compare the µ of data you gathered to an already existing µ (e.g. from literature)
    - **Independent t-test**. Used to compare the µ´s of two groups you observed
    - **Paired samples t-test**. Used to compare the µ´s of data you gathered within one group, but at two different moments (e.g. µ heart rate before and µ after drinking coffee). ¡This requires the same people in both groups!
- Here, you have to set clear one-sided or two-sided hypotheses to choose the right settings and interpret results correctly. *In case of one-sided testing, only half of the chosen p-value is used*. Jamovi automatically changes the p when you tick one-sided; in SPSS you would have to read to column ´one-sided´.
- You find T-tests in Jamovi under Analyses > T-tests.
- To check data assumptions, in Jamovi tick *Homogeneity test* (Levene´s) and *Normality test* (Shapiro-Wilk).
    - If both p-values are **higher** than 0.05 you can move forward with a normal T-test.
    - If either one of them is violated or both are, you have to tick the *Mann-Whitney U* option and then continue. (The *Mann-Whitney U* uses medians instead of means)
- For effect size, tick *Effect Size* under *Additional Statistics*. This will add Cohen´s d (applies when doing a normal t-test) and Rank biserial correlation (applies when doing the Mann-Whitney U).

Normality Test (Shapiro-Wilk) ✓

	W	p
perf_quantity	0.979	0.094

Note. A low p-value suggests a violation of the assumption of normality

Homogeneity of Variances Test (Levene's) ✓

	F	df	df2	p
perf_quantity	0.256	1	104	0.614

Note. A low p-value suggests a violation of the assumption of equal variances

Independent Samples T-Test X

		Statistic	df	p		Effect Size
perf_quantity	Student's t	1.43	104	0.157	Cohen's d	0.383
					Rank biserial correlation	−0.206

Note. H_a $\mu_2 \neq \mu_1$

ANOVA

- ANOVA is used to compare means (μ´s) of **three or more groups** for a continuous dependent variable, to see if there is a significant difference between those groups. You have to know 2 options for ANOVA, for which the choice is based on Levene´s test of homogeneity (as a data assumption).
  - **ANOVA Fisher´s test**. Used if homogeneity is met (so: Levene´s $p > 0.05$)
  - **ANOVA Welch test**. Used if homogeneity is **not** met (so: Levene´s $p < 0.05$)
- You find ANOVA in Jamovi under Analyses > ANOVA > One-way ANOVA.
- To check homogeneity data assumption, in Jamovi tick *Homogeneity test* (Levene´s) within the ANOVA test

However…. if a significant difference is found, we still have to see **between which groups**, for which we use post-hoc tests. These post-hoc tests show p-values for specific pairs of groups so you can report outcomes more precisely. To also report which groups scores higher than another, use desriptives to see groups´ means. Post-hoc options are:

- For an ANOVA Welch test we use the Games-Howell
- For an ANOVA Fisher´s test we use the Bonferroni or Tukey, for which the choice depends on acceptable risks:
  - Bonferroni is quite strict, which means a lower chance of a type I error, yet a bigger chance of a type II error
  - Tukey is less strict, which means a higher chance of a type I error, yet a smaller chance of a type II error. Jamovi only has Tukey.

Tukey Post-Hoc Test – perf_quantity

		1	2	3	4	5
1	Mean difference	—	0.365	1.132	–0.439	–1.136
	p-value	—	0.734	0.156	0.742	0.324
2	Mean difference		—	0.767	–0.804	–1.501
	p-value		—	0.486	0.117	0.081
3	Mean difference			—	–1.571	–2.268
	p-value			—	0.024	0.014
4	Mean difference				—	–0.697
	p-value				—	0.789
5	Mean difference					—
	p-value					—

Assumption Checks

Homogeneity of Variances Test (Levene's)

	F	df1	df2	p
perf_quantity	1.68	4	101	0.160

One-Way ANOVA

		F	df1	df2	p
perf_quantity	~~Welch's~~	~~8.30~~	~~4~~	~~23.0~~	~~<.001~~
	Fisher's	4.18	4	101	0.004

NOTES

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